

Who

Data Set

Variables and Model

Project Check In 3

Who

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Data Set

- https://data.austintexas.gov/Environment/Water-Quality-Sampling-Data/5tye-7ray/about_data
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- Austin Water Quality Sampling Data
- Each row is a water quality sample by parameter, date, and location in Austin, TX. We are primarily interested in variables watershed, site_type, and result to understand the distribution of water quality across Austin.
- There are 1,475,965 rows and 24 columns in the original data set. Because there is a lot of data, we have limited the data set to bacteria/pathogen samples. The data set 'water' contains 34893 rows and 24 columns.

```
# Import 2024 Austin water quality sampling data set
water <- read.csv('https://raw.githubusercontent.com/Amandawlee/sds322_project/ref
s/heads/main/Water_Quality_Sampling_Data.csv')

# Number of rows and columns in 'water'
dim(water)
```

```
## [1] 34893    24
```

Variables and Model

- **Model 1 (Amanda):** Predict if a bacteria/pathogen concentration sample is considered safe or unsafe.
 - **Model Type:** Decision Tree

- **Outcome Variable and Type:** safeness (Categorical)
- **Predictors and Types:** site_type (Categorical), parameter (Categorical), result (Numerical)
- **Model 2 (Nabil):** Predict bacteria concentrations using time of day, type of water source, and site type.
 - **Model Type:** Linear Regression
 - **Outcome Variable and Type:** Result , Numeric
 - **Predictors and Types:** sample_date (Numeric), Medium (Categorical), and site_type (Categorical).